

Complete Problem Landscape of Hybrid AI Architectures (TSM)

This document provides a comprehensive overview of the challenges involved in designing a hybrid sequence model combining Transformer (attention), Mamba (state-space), and Hyena (convolution) architectures.

Tier 1: Core Problems

Representation Mismatch

Different architectures use different internal representations, making it difficult to combine their outputs effectively.

Fusion Problem

Determining how to combine outputs from multiple modules without causing dominance or collapse.

Training Instability

Different learning speeds and gradient conflicts lead to unstable training.

Computational Complexity

Running multiple architectures increases compute and memory requirements.

Over-engineering

Added complexity may not justify performance improvements over simpler models.

Interaction Problem

Designing intelligent cooperation between modules is the core challenge.

Tier 2: Advanced Research Problems

Credit Assignment Problem

Difficult to determine which module contributed to the output.

Module Dominance

One module may dominate while others become redundant.

Temporal Misalignment

Different modules interpret sequence timing differently.

Information Redundancy

Multiple modules may capture overlapping information.

Gating Complexity

Dynamic routing between modules is complex and context-dependent.

Scaling Behavior

Performance at large scale is uncertain.

Evaluation Problem

Hard to measure improvements across multiple dimensions.

Implementation Complexity

Difficult to build, debug, and optimize in practice.

Conclusion

Hybrid architectures like TSM represent a promising direction in AI, but solving these challenges is essential for achieving practical and scalable systems.